MIL-T-22939B(OS)
6 October 1972
Superseding
MIL-T-22939A(OS)
28 February 1968

MILITARY SPECIFICATION

TEST SET, TORPEDO, MK 393 MOD 0

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

2. SCOPE

1.1 This specification describes the requirements for the procurement of Torpedo Test Set Mk 393 Mod 0, LD 539994, which is used to check the communications link of Torpedo Mk 37 Mod 2. The test set provides all necessary control signals required for testing the torpedo's communications system. The test set is enclosed in a portable, waterproof combination case.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Methods of Preservation
MIL-T-945	Test Equipment for Use with Electronic Equipment; General Specification

STANDARDS

Federal

Federal Test Method Standard No. 101	Preservation, Packaging, and Packing Materials, Test Procedures

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 1355

Source: https://assist.dla.mil -- Downloaded: 202610-19T14:56Z
Check the source to verify that this is the current version before use.

MIL- STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base
MIL-STD-810	Environmental Test Methods

DRAWINGS

Naval Ordnance Systems Command

NAVORD LD 539994	Torpedo Test Set Mk 393 Mod 0 and all Parts and Subassemblies listed thereon
NAVORD Drawing 2212066	Torpedo Test Set Mk 393 Mod 0 Assembly

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

National Bureau of Standards Publications

Handbook H28	Screw Thread Standards for the Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. Torpedo Test Set Mk 393 Mod 0 (LD 539994) is primarily intended for use in checking the communication link of Torpedo Mk 37 Mod 2. The unit, which applies command signals to the torpedo through the weapon's wire guidance system, may be used during the actual run of the torpedo or in pre-run and post-run tests. The set incorporates the following functions:

(a) It provides a 42 volt (±) dc signal for the activation of the torpedo steering controller.

(b) It provides a 400 Hertz (Hz), 200 volt signal for the activation of the command search function.

(c) It receives signals from the 60 Hz oscillator, located in the torpedo, which indicates the torpedo's status.

Source: https://assist.dla.mil -- Downloaded: 202610-19T14:56Z
Check the source to verify that this is the current version before use.

(d) It includes a line simulator network that provides an impedance equivalent to the impedance of the torpedo wire coil, which may be inserted in the communication circuit when required.

The test set, which requires a power source of 115 $^{+10}_{-5}$ volts rms 400 Hz ±5 percent for operation, is enclosed in a portable watertight combination case. Dimensionally, the unit (including combination case) i. 14 inches long, 9 inches wide, and 8½ inches high.

3.2 General requirements

3.2.1 Preproduction sample. Unless othewise specified in the contract or order, preproduction samples of the test set shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing test sets that comply with the technical requirements of the contract. See 4.3.1.

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the test set shall conform strictly to the detailed requirements indicated on the drawings and specifications in NAVORD LD 539994.

3.2.3 Finishes, coating. construction, and dimensions. Finishes, coatings, construction, and dimensions for the test set shall be strictly in accordance with the applicable drawings, specifications, and publications listed in NAVORD LD 539994.

3.2.4 Moisture, fungus, and corrosion. The test set shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified in the drawings and specifications listed in NAVORD LD 539994.

3.2.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in NAVORD LD 539994.

3.2.6 Threads. Unless othewise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.7 Conflicting requirements. in the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.3 Definitions. Definitions applicable to the environmental requirements of this specification are defined in MIL-T-945.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

3.4 Performance and product characteristics. The test set shall meet the following performance requirements and product characteristics:

3.4.1 The test set shall provide the following voltages across a terminating load of 3000 ±5 percent ohms with the line simulator out of the test circuit:

3.4.1.1 Command steps: +40 and -40 volts dc minimum.

3.4.1.2 Command search: 185 volts rms minimum, 400 Hz.

3.4.2 The test set shall provide the following voltages measured across a terminating load of 1500 25 percent ohms with the line simulator network in the circuit:

3.4.2.1 Command steps: +15 and -15 volts dc minimum.

3.4.2.2 Command search: 4.5 volts rms minimum, 400 Hz.

3.4.3 Right step commands (+dc voltage) shall be additive on the command step counter.

3.4.4 Left step commands (-de voltage) shall be subtractive on the command step counter.

3.4.5 With the line simulator network out of the test circuit, the status light shall be actuated by application of 5.15 ±3.65 volts rms, 60 Hz to the test set output terminals.

3.4.6 With the line simulator network in the test circuit, the status light shall be actuated by application of 10.2 ±8.1 volts rms, 60 Hz to the test set output terminals.

3.4.7 Operating life. The test set shall be capable of meeting operating life requirements as specified in MIL-T-945 under section titled “Life.”

3.5 Environmental requirements. The test set shall be capable of meeting the performance requirements and product characteristics specified herein after or while being subjected to the following conditions, when tested as defined in Section 4,

3.5.1 Vibration. The test set shall be capable of meeting all the performance requirements and product characteristics of this specification after being subjected to vibration in accordance with Test I of MIL-T-945.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

3.5.2 Shock. The test set shall be capable of meeting shock requirements as specified in Section 4 of MIL-T-945, with nine hammer blows, each equivalent to an acceleration of 50 gravity-units, applied.

3.5.3 Temperature. The test set shall be capable of meeting temperature requirements as specified in Section 4 of MIL-T-945. Operating temperature limits shall range from +28°F to +130°F.

3.5.4 Humidity. The test set shall be capable of meeting humidity requirements as specified in Section 4 of MIL-T-945.

3.5.5 Tropical environment. The test set shall be capable of meeting fungus requirements as specified in Section 4 of MIL-T-945.

3.5.6 Marine environment. The test set shall be tested by exposure to the weather conditions described in 4.5.6. At the conclusion of the test, the test set shall meet the performance requirements specified in 3.4.

3.5.7 Serviceability requirements, The test set shall show no damage, after being subjected to conditions defined below, when tested as specified in Section 4.

3.5.7.1 Low order repetitive shock (vibration). The packed test set, after being subjected to the tests specified in 4.5.8, shall show no damage.

3.5.7.2 Vibration (sinusoidal motion) test. The packed test set, after being subjected to the test specified in 4.5.9, shall show no damage.

3.5.7.3 Free drop test (shock). The packed test set, after being subjected to the tests specified in 4.5.10, shall show no damage.

3.5.7.4 Leak test. The packed test set, after being subjected to the test specified in 4.5.11, shall show no damage.

3.6 Cleanup. After final assembly all test sets shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.7 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval,

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as othewise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the test set shall be classified as follows:

(a) Preproduction inspection (paragraph 4.5)

(b) Quality conformance inspection (paragraph 4.4)

(c) Periodic production inspection (paragraph 4.5)

4.3 Sampling

4.3.1 Preproduction. A preproduction sample of five test sets, manufactured in accordance with 3.2.1, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the test set by the supplier, prior to the approval of the preproduction sample, shall be at the supplier’s risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate “inspection lot” and may differ from the quantity specified in the total order or contract. In any event, all units of a single “inspection lot” shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five test sets will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The test sets shall be considered satisfactory if they meet all specifications and pertinent drawings requirements and withstand the tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Quality conformance inspection. Each test set delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection.

4.4.1 Test condition. Unless otherwise specified, the test set shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Temperature. Room ambient 65°F to 95°F (18°C to 35°C).

4.4.1.2 Altitude. Normal ground.

4.4.1.3 Vibration. None.

4.4.1.4 Humidity. Room ambient to 95 percent relative maximum.

4.4.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the specified tests.

4.4.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.4.3.1 A 500-volt megohmmeter with full scale of not less than 10 megohms.

4.4.3.2 A voltmeter capable of measuring 400 Hz sinusoidal voltage to an accuracy of 1 percent full scale (10, 30, 300 volt f.s. range.) with an input impedance of not less than 10 MΩ (hp (hp 400H or equal).

4.4.3.3 An alternating current power source, single phase variable 100-130 volts rms, 400 Hz ±5 percent, total harmonic content 5 percent maximum, single harmonic not to exceed 3 percent.

4.4.3.4 A dc decoupled power supply, variable 6 to 20 volts rms, 60 Hz ±5 percent.

4.4.3.5 A voltmeter capable of measuring dc voltage to an accuracy of ±3 percent f.s. within the range of 0 to 100 volts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

4.4.3.6 A 3000 ohms ±5 percent, 10 watts load resistor.

4.4.3.7 A 1500 ±5 percent ohm, 10 watts load resistor.

4.4.3.8 An ac voltmeter capable of measuring 115 volts rms 400 Hz to an accuracy of ±0.1 percent of input +25 μv (Fluke Model 883 or equal).

4.4.4 Test procedure

NOTE: The command step and command search switches (identified on test set front panel as STEP SWITCH and CS SWITCH, respectively) are momentary contact switches having a normal center position. The CS SWITCH is activated by an up movement to the ON position, while the STEP SWITCH is activated by a left or right movement to either the “L” or “R” position.

4.4.4.1 The following test procedure shall be performed in the order indicated:

4.4.4.2 Turn POWER OFF switch off (down) and LINE SIMULATOR switch to OUT. Leave all other switches in their normal position.

4.4.4.3 Terminate the test set output circuit with the 3k ± 5 percent ohms, 10 watts load resistor.

4.4.4.4 Connect the power source of 4.4.3.3 to the appropriate input receptacle of the test set. Adjust the supply voltage to 110 vrms, as measured on the meter of 4.4.3.8.

4.4.4.5 Turn POWER OFF switch to on (up). The POWER lamp shall light,

4.4.4.6 Remove fuse. The POWER lamp shall go off.

4.4.4.7 Replace fuse. The POWER lamp shall light.

4.4.4.8 Turn CS SWITCH to ON and hold.

4.4.4.8.1 using the meter of 4.4.3.2 measure 182 volts rms minimum, 400 Hz, across the load resistor.

4.4.4.8.2 Release CS SWITCH.

4.4.4.9 Turn LINE SIMULATOR switch to IN, and replace the 3k ±5 percent 10 watts load resistor with the 1.5k ±5 percent 10 watts load resistor. resistor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

4.4.4.10 Turn CS SWITCH to ON and hold.

4.4.4.10.1 Using the meter of 4.4.3.2 measure 4.5 volts rms, 400 Hz, minimum across the load resistor.

4.4.4.10.2 Release CS SWITCH.

4.4.4.11 Turn POWER OFF switch off (down).

4.4.4.12 All lamps shall be off.

4.4.4.13 Turn LINE SIMULATOR switch to OUT.

4.4.4.14 Remove the 1.5k ±5 percent ohm load resistor from the test set output terminals and replace it with the 3k ±5 percent ohm load resistor. Connect the power supply of 4.4.3.4 and the voltmeter of 4.4.3.2 across the load resistor. Do not energize the 60 Hz supply at this time.

4.4.4.15 Turn POWER OFF switch to ON (up). The POWER lamp shall light,

4.4.4.16 Energize the 60 Hz supply connected across the load resistor. Slowly increase the ac voltage across the load resistor until the STATUS lamp lights. STATUS lamp actuation shall occur between 1.5 and 8.8 vac, as indicated on the meter of 4.4.3.2.

4.4.4.16.1 Connect the dc voltmeter of 4.4.3.5 across the load resistor.

4.4.4.16.2 Measure less than one volt dc across the load resistor.

4.4.4.17 Set the STEP SWITCH to “L” and hold.

4.4.4.17.1 The LINE CURRENT meter shall read -I2 milliamperes dc minimum.

4.4.4.17.2 Measure -40 volts dc minimum across the load resistor.

4.4.4.18 Release STEP SWITCH.

4.4.4. 18.1 The BEARING dial (command gyro angle) shall read 358° ±½°.

4.4.4.18.2 The LINE CURRENT meter shall read O milliamperes.

4.4.4.19 Set STEP SWITCH to “R” and hold.

4.4.4. 19.1 The LINE CURRENT meter shall read +12 milliamperes dc minimum.

4.4.4.19.2 Measure +40 volts dc minimum across the load resistor,

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T14:56Z
Check the source to verify that this is the current version before use.

4.4.4.20 Release STEP SWITCH.

4.4.4.20.1 The BEARING dial shall read 000° ±½°.

4.4.4.20.2 The LINE CURRENT meter shall read O milliamperes.

4.4.4.21 Reduce the 60 Hz signal across the load resistor to O volts. Disconnect the dc voltmeter. Replace the 3k ±5 percent ohm, 10 watt load resistor with the 1.5k ±5 percent ohm, 10 watt load resistor. Turn LINE SIMULATOR switch to IN.

4.4.4.22 Slowly increase the ac voltage across the load resistor until the STATUS lamp lights. STATUS lamp actuation shall occur between 2.1 and 18.3 volts rms as indicated on the meter of 4.4.3.2.

4.4.4.22.1 Connect the dc voltmeter of 4.4.3.5 across the load resistor.

4.4.4.23 Turn STEP SWITCH to “R” and hold.

4.4.4.23.1 The LINE CURRENT meter shall read +9.0 milliamperes dc minimum.

4.4.4.23.2 Measure +15 volts dc minimum across the load resistor.

4.4.4.24 Release STEP SWITCH.

4.4.4.24.1 The BEARING dial shall read 002° ±½°.

4.4.4.24.2 The LINE CURRENT meter shall read 0 milliamperes.

4.4.4.25 Turn STEP SWITCH to “L” and hold.

4.4.4.25.1 The LINE CURRENT meter shall read -9.0 milliamperes dc minimum.

4.4.4.25.2 Measure -15 volts dc minimum across the load resistor.

4.4.4.26 Release STEP SWITCH.

4.4.4.26.1 The BEARING dial shall read 000° ±½°

4.4.4.26.2 The LINE CURRENT meter shall read 0 milliamperes.

4.4.4.27 Disconnect the 60 Hz power source from the test set output terminals. Replace the 1.5K ±5 percent ohm, 10 watt load resistor with the 3k 25 percent ohm, 10 watt load resistor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

4.4.4.28 Turn LINE SIMULATOR switch to OUT.

4.4.4.29 Turn CS SWITCH to ON and hold.

4.4.4.29.1 Measure 185 volts rms minimum, 400 Hz across the load resistor.

4.4.4.30 Release CS SWITCH.

4.4.4.31 Turn POWER OFF switch off (down).

4.4.4.32 Disconnect input power from test set.

4.4.4.33 Disconnect load resistor and all auxiliary test equipment from test set.

4.4.4.34 With all switches in the normal position, the measured insulation resistance between the chassis and the following points shall be a minimum of 10 megohms: line-line ground-both sides of input power receptacle. Close all the switches and repeat the above insulation resistance tests.

4.4.4.35 Visual and dimensional examinations. Each test set shall be examined to detemine conformance with the requirements for workmanship and dimensions.

4.4.4.36 Packaging packing. and marking. Examine the packaging, packing, and marking to determine conformance with the requirements of Section 5.

4.5 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.5.1 Vibration. The test set, with cover closed and secured, shall be vibration tested in accordance with Test I of MIL-T-945. Thereafter, the test set shall meet the performance requirements specified in 3.4.

4.5.2 Shock. The test set shall be firmly attached to the table of the shock-testing machine, and subjected to the shock test procedure defined in MIL-T-945. After each axis of shock, the test set shall perform in accordance with the requirements of this specification.

4.5.3 Temperature. The test set shall be tested in accordance with temperature test procedures in MIL-T-945. Operating temperature shall be as specified in 3.5.3. At the conclusion of the test, the test set shall meet the performance requirements specified in 3.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

4.5.4 Humidity. The test set shall be tested in accordance with humidity test procedures in MIL-T-945. At the conclusion of the test, the test set shall meet the performance requirements specified in 3.4.

4.5.5 Trcpical environment. The test set. shall be fungus tested in accordance with MIL-T-945. At the conclusion of the test, the test set shall meet the performance requirements specified in 3.4.

4.5.6 Marine environment. The test set, with cover closed and secured, shall be exposed to the sea atmosphere as encountered on the weather deck of a ship for 10 ±1 days, protected from rain, spray, or snow by a shelter which permits free air circulation around the test set. At the conclusion of the test, the test set shall meet the performance requirements specified in 3.4.

4.5.7 Operating life. The test set shall be energized for a period of 200 hours under maximum operating temperature and humidity conditions as follows: +130°F at 95 percent relative humidity. At the conclusion of the test, the test set shall meet the performance requirements specified in 3.4.

4.5.8 Low order repetitive shock (vibration). his test shall be run prior to running other standard shock tests. This test shall be run in accordance witn Method 278 of Federal lest Metnod Standard No. 101 with the following clarification:

(a) If circular input motion is used, table frequency shall be adjusted to assure that one edge of the container leaves the table on each cycle by 3/16 inch.

(b) If vertical linear input motion is used, all points of the container shall leave the table by 1/16 inch at some instant during each cycle.

(c) If neither of the above is attainable with equipment at hand, the cable shall be operated at 270 ±2 rpm for the stipulated time period.

4.5.9 Vibration (sinusoidal motion) test. This test shall be run in accordance with Method 5020 of Federal Test Method Standard No. 101 with the following exception:

(a) The frequency range of 5 to 60 Hz for a 10-minute sweep to determine resonance will be maintained in lieu of 105 minutes as stated under Method 5020.

4.5.10 Free-drop tests (shock). These tests shall be conducted to requirements as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

(a) Ambient temperature test. This test shall be canducts. at the temperature of 21.1°C (70°F) from the heights stipulated in Table I, drop in succession on corners 5-1-2, 6-3-4, 5-2-3, and 6-1-4. Then drop on face 3 and face 5 (see Figure 1). Suitable test rigging is described in Method 216 of Federal Test Method Standard No. 101.

(b) High temperature test. This test shall be conducted in accordance with the requirements of MIL-STD-810, Method 501, Procedure I. From the height stipulated in Table I, drop in succession on comers 5-1-2, 6-3-4, 5-2-3, and 6-1-4. Then drop on face 3 and face 5 (see Figure 1). Suitable test rigging is described in Method 216 of Federal Test Method Standard No. 101.

(c) Low temperature test. This test shall be conducted In accordance with the requirements of MIL-STD-810, Method 502. From the height stipulated in Table I, drop in succession on comers 5-1-2, 6-3-4, 5-2-3, and 6-1-4. Then drop on face 3 and face 5 (see Figure 1). Suitable test rigging is described in Method 216 of Federal Test Method Standard No. 101.

4.5.11 Leak test. The hot water technique shall be conducted in accordance with Method 5009 of Federal Test Method Standard No. 101.

5. PREPARATION FOR DELIVERY

5.1 Presentation and packaging. The levels of preservation and packaging shall be as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit packaging. The test set shall be packaged in accordance with Method IID of MIL-P-116. The unit package shall contain one test set.

5.1.1.2 Intermediate package. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C

5.1.3.1 Unit packaging. The test set shall be packaged in accordance with Method III of MIL-P-116. The unit package shall contain one test set.

5.1.3.2 Intermediate package. Not applicable.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Number of packages per pack. The number of unit packages per pack, for Level C, shall be as specified in the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

TABLE I

Shipping Container Shock Test Selector

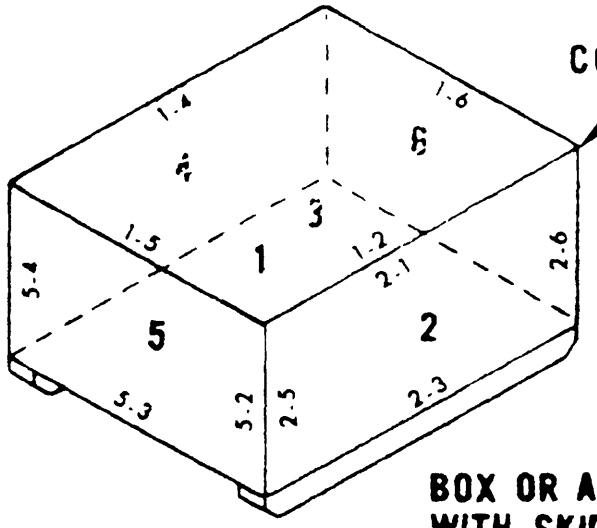
Container Gross Wt. W, lbs./1	Container Length, L, Ins. /1	Type Test	Domestic Only /2		Universal	
			Ht. of Drop ins.	Impact Velocity fps.	Ht of Drop ins.	Impact Velocity fps.
W≤25	Any	Free Drop	30	N.R./3	48	N.R.
25<W≤50	L≤36	Free Drop	30	N.R.	36	N.R.
25<W≤50	L>36	Rotation & Impact	30	7	36	12½
50<W≤150	L≤36	Free Drop	14	N.R.	30	N.R.
50<W≤150	L>36	Rotation & Impact	27	7	30	10
150<W≤500	Any	Rotation & Impact	19	7	24	10
500<W≤4000	Any	Rotation & Impact	12	7	18	10
W>4000	Any	Rotation & Impact	9	7	12	10

/1 Using both column to determine the applicable test conditions.

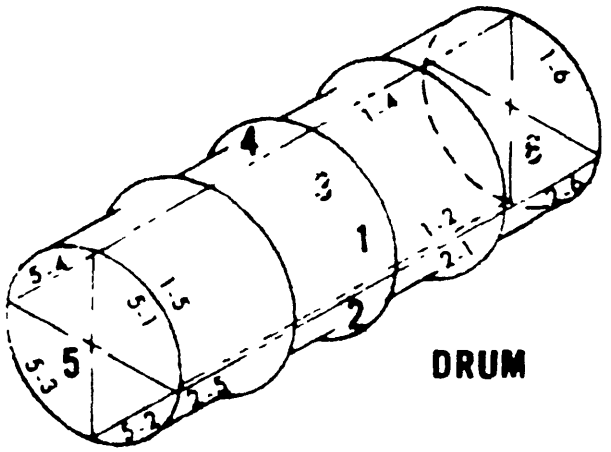
/2 This column may not be used if the contents are AEDA. Containers for AEDA developed under direct military control shall be designed and tested to the universal level. Bulk materials and commercially developed AEDA must be shipped in containers meeting DOT Regulations, including all performance test requirements, unless specific waiver be granted by NAVORD. Where such waiver exists, the shipper may assume that the waiver has been coordinated with appropriate regulatory agencies.

/3 N.R. means not required.

Source: https://assist.dla.mil -- Downloaded: 202610-19T14:56Z
Check the source to verify that this is the current version before use.



BOX OR ANY CONTAINER WITH SKIDS



DRUM

DEFINITION OF FACES, EDGES & CORNERS

Figure 1

Source: https://assist.dla.mil -- Downloaded: 20261O-19T14:56Z
Check the source to verify that this is the currentversion before use.

5.2.2 Level A

5.2.2.1 Unit pack. The test set shall be packed into a reuseable, metal container in accordance with 5.1.1.1.

5.2.2.2 Palletizing. When specified, Level A packs shall be palletized in accordance with MIL-STD-147

5.2.3 Level B. Not applicable.

5.2.4 Level C. the test set shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Levels A or C cushioning. The unit package shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the test set will meet all the requirements of. Section 3 after being tested in accordance with 4.5.8, 4.5.9 and 4.5.10. At the time of packaging or packing, the moisture content of the cushioning material shall be no more t-ban 5 percent.

5.4 Marking

5.4.1 Special markings. None , unless otherwise specified.

5.4.2 Normal markings. In addition to the markings required by the contract or order unit packages and shipping containes shall be marked in accordance with the requiements of MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended use. Torpedo Test Set Mk 393 Mod 0 is intended for use in the operational checking of Torpedo Mk 37 Mod 2.

6.2 Ordering Data. Procurement documents should specify the following:

6.2.1 Procurement requirements

(a) Title, number, and date of this specification.

(b) Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletization (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

(f) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.2.1 Contract data requirements. Test reports associated with any of tune tests covered in paragraph 4.5 of this specification will be prepared and delivered by the contractor only when listed on a Contract Data Requirements List (DD Form 1423) incorporated into the contract. In those instances, the reports shall be prepared in accordance with the following Data Item Description (DID) cited by number in Block 4 of the DD Form 1423:

DID#	TITLE
DI-T-2072	Report, Test

Custodian:
Navy - OS

Preparing Activity
Navy - OS
Project Number 1355-N296

* U.S. GOVERNMENT PRINTING OFFICE 1972 -714-910/387

Source: https://assist.dla.mil -- Downloaded: 20261O-19T14:56Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-22939B (OS) Test Set, Torpedo, Mk 393 MOD 0

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

S/N-0102-014-1801 C-28284

Source: https://assist.dla.mil -- Downloaded: 202610-19T14:56Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

Naval Ordnance Station
Indian Head, MD 20640

OFFICIAL BUSINESS

Commanding Officer
Naval Ordnance Station
Standardization Division (QA3)
Indian Head, MD 20640

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T14:56Z
Check the source to verify that this is the current version before use.